IGCSE 0607 Practice Questions — E10.3 — Combined Questions & Markscheme

Discrete and continuous data

Total Questions: 18 — **Total Marks:** 93 Marks

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1 A weather station records the following data each day.
A: The number of times lightning strikes.
B: The maximum wind speed.
C: The volume of water collected in a rain gauge.
D: The number of days it snows in a year.

(a) Write down the letters of the variables that represent discrete data.

..... **[1]**

(b) Write down the letters of the variables that represent continuous data.

..... **[1]**

2 A scientist records different types of weather data.
State whether each of the following types of data is discrete or continuous.

(a) The daily maximum temperature.

..... **[1]**

(b) The number of rainy days in a month.

..... **[1]**

(c) The amount of rainfall in one hour.

..... **[1]**

3 Statistical data can be classified as either discrete or continuous.

(a) State whether each of the following variables is discrete or continuous data.

(i) The number of cars in a car park.

..... **[1]**

(ii) The temperature of a cup of coffee.

..... **[1]**

(iii) The time it takes to complete a jigsaw puzzle.

..... **[1]**

(iv) The number of keys on a piano.

..... **[1]**

4 A sports analyst collects data during a football match.

(a) State whether the number of goals scored in the match is discrete or continuous.

..... **[1]**

(b) State whether the mass of the football is discrete or continuous.

..... **[1]**

(c) State whether the time taken to complete the match is discrete or continuous.

..... **[1]**

(d) State whether the temperature on the pitch is discrete or continuous.

..... **[1]**

5 A student collects data for a statistics project.
For each of the variables below, state whether the data collected is discrete or continuous.

(a) The number of goals scored in a football match.

..... **[1]**

(b) The mass of a football.

..... **[1]**

(c) The volume of water in a swimming pool.

..... **[1]**

(d) The number of people swimming in a pool.

..... **[1]**

6 An environmental science club records four different types of statistical data during a field trip to a local river.

(a) State whether each of the following sets of data is discrete or continuous:

(i) the number of fish counted in each net sample,

..... **[1]**

(ii) the water temperature, in degrees Celsius,

..... **[1]**

(iii) the width of the river, in metres,

..... **[1]**

(iv) the shoe size of each student attending the field trip.

..... **[1]**

(b) Explain why the width of the river is classified as continuous data.

..... **[1]**

(c) A student measures the widths, w metres, of the river at 10 points and groups the continuous data into intervals of width 2 metres starting at 4 metres.
Explain why writing the first two intervals as $4 \leq w \leq 6$ and $6 \leq w \leq 8$ is not suitable, and write down the correct inequality notation for the first interval.

.....

..... **[2]**

7 A hardware store sells nails in large boxes.

(a) The mass of one box of nails is measured as 40 kg, correct to the nearest 5 kg.
Mass is a continuous variable.
Find the lower bound and the upper bound for the mass of the box.

Lower bound = ... , Upper bound = ... **[2]**

(b) The number of nails in the box is 1250, correct to the nearest 5.
The number of nails is a discrete variable.
Write down the smallest possible number of nails and the largest possible number of nails in the box.

Smallest possible number = ... , Largest possible number = ...
[2]

8 Two sets of data are collected during a fitness test.
Data Set A is the exact time, t seconds, taken by athletes to run a race.
Data Set B is the number of push-ups, n, completed by the athletes.

(a) Data Set A is grouped into classes such as $10 \leq t < 12$.
Data Set B is grouped into classes such as $10 - 19$.
Explain why the classes for Data Set A use inequalities while the classes for Data Set B use a hyphen.

.....

..... **[2]**

(b) A student decides to draw a histogram to represent Data Set B. One of the classes is $20 - 29$.

(i) State the lower and upper class boundaries that must be used for this class on the horizontal axis of the histogram.

Lower = ... , Upper = ... **[1]**

(ii) Calculate the class width used for the frequency density calculation.

..... **[1]**

(c) Continuous data can theoretically take any value within a range.
Give an example of a continuous variable that is practically recorded as discrete data in everyday life.

.....

..... **[1]**

9 A factory manufactures metal rods and packs them into boxes. For each box, two variables are recorded:

- k: the number of rods in the box.
- m: the total mass, in kilograms, of the rods in the box.

(a) State whether k is discrete or continuous, and state whether m is discrete or continuous.

k is ..... , m is ..... **[2]**

(b) Each rod has a mass of 2.0 kg, correct to 1 decimal place.
A box of k rods has a total mass of 20.0 kg, correct to 1 decimal place.
By considering the lower and upper bounds of the continuous mass measurements, show that $k = 10$.

[4]

(c) The factory inspects 50 boxes and records the number of defective rods, d, in each box and the length, L cm, of a sample of 50 rods.
The results are shown in the two tables below.

Defective (d)	Frequency
0	24
1	15
2	7
3	3
4	1

Length (L cm)	Frequency
$49.6 \leq L < 49.8$	8
$49.8 \leq L < 50.0$	18
$50.0 \leq L < 50.2$	17
$50.2 \leq L < 50.4$	7

(i) Find the exact mean number of defective rods per box, and explain why this value is exact whereas the mean length calculated from the second table is only an estimate.

..... **[3]**

(ii) A rod is accepted if its length satisfies $49.8 \leq L < 50.2$.
Calculate the percentage of the 50 sampled rods that are accepted, and explain why it is not possible from the table to determine the exact number of rods with length greater than 50.1 cm.

.....

..... **[2]**

10 A botanist studies a new species of plant in a greenhouse.
She records the height, h cm, of each plant and the total number of leaves on each plant.

(a) State whether the height of a plant is discrete or continuous data.

..... **[1]**

(b) The botanist wants to group the heights of the plants into a frequency table.
She suggests using the following class intervals:

$$10 - 19 \qquad 20 - 29 \qquad 30 - 39$$

Explain why these class intervals are not appropriate for the height of the plants.

.....

..... **[2]**

(c) State whether the number of leaves on a plant is discrete or continuous data.

..... **[1]**

11 A student collects data about a local park for a science project.

(a) Explain the mathematical difference between discrete and continuous data.

.....

.....

..... **[2]**

(b) The student measures the lengths of 50 leaves to the nearest millimetre.

(i) State whether the true length of a leaf is discrete or continuous.

..... **[1]**

(ii) Explain why measuring the lengths to the nearest millimetre does not make the variable discrete.

.....

..... **[1]**

12 A farmer weighs 60 apples.
The masses, m grams, are recorded in the table.

Mass (m grams)	**Frequency**
$80 < m \leq 100$	12
$100 < m \leq 120$	26
$120 < m \leq 140$	15
$140 < m \leq 160$	7

(a) State whether the mass of an apple is discrete or continuous data.

..... **[1]**

(b) Calculate an estimate of the mean mass.

..... g **[3]**

(c) Explain why your answer to part (b) is an estimate.

..... **[1]**

13 A scientist measures the lengths of 50 worms and counts the number of spots on each worm.
The frequency table shows the data collected for the number of spots.

Number of spots	$0-4$	$5-9$	$10-14$	$15-19$
Frequency	12	18	14	6

(a) State which of the two variables, length or number of spots, is continuous.

..... **[1]**

(b) The length of one worm is recorded as 14.5 cm, correct to the nearest millimetre.

(i) Write down the lower bound of this length.

..... *cm* **[1]**

(ii) Write down the upper bound of this length.

..... *cm* **[1]**

(c) Using the frequency table, write down the class width for the $5-9$ class.

..... **[2]**

14 A factory tests a sample of 40 light bulbs.
It records the lifetime, t hours, of each bulb.

(a) State whether the lifetime of a light bulb is discrete or continuous.

..... **[1]**

(b) The results are shown in the table.

Lifetime (t hours)	$100 < t \leq 200$	$200 < t \leq 300$	$300 < t \leq 400$	$400 < t \leq 500$
Frequency	4	14	16	6

Calculate an estimate of the mean lifetime.

..... hours **[4]**

15 A sports analyst collects data on 40 athletes competing in an athletics meet. Two variables are recorded for each athlete:

- N: the number of training sessions attended in the previous month.
- T: the time taken, in seconds, to complete a 400 m sprint.

The two tables below summarise the results.

Sessions (N)	Frequency
$10 \leq N \leq 12$	6
$13 \leq N \leq 15$	14
$16 \leq N \leq 18$	12
$19 \leq N \leq 21$	8

Time (T seconds)	Frequency
$48 \leq T < 50$	5
$50 \leq T < 52$	15
$52 \leq T < 54$	13
$54 \leq T < 56$	7

(a) Explain how the inequality notation used for the class intervals in the two tables reflects the difference between discrete data and continuous data.

.....

..... **[2]**

(b) Calculate an estimate of the mean number of training sessions attended, N, and an estimate of the mean sprint time, T.

Mean of N = ... , Mean of T = ... **[4]**

(c) **(i)** A student says: "The estimated mean of N is 15.65, which is impossible because N is discrete data and an athlete cannot attend 15.65 training sessions." Explain whether the student is correct about the validity of the mean.

..... **[2]**

(ii) State a suitable type of statistical diagram to represent the distribution of discrete data N and a suitable type of statistical diagram to represent the distribution of continuous data T.

Diagram for $N =$ ..... , Diagram for $T =$ ..... **[2]**

16 Data is collected about the members of an athletics club.

(a) State whether each of the following variables is discrete or continuous.

(i) The number of members in the club.

..... **[1]**

(ii) The exact distance a member jumps in the long jump.

..... **[1]**

(b) The shoe sizes of the members are recorded.
Shoe sizes are available in sizes such as $7, 7.5, 8, 8.5, 9$.
A student claims that because these shoe sizes include decimal values, the data is continuous.
Explain why the student is incorrect.

.....

.....

..... **[2]**

17 Measurements and quantities in statistics are defined by their ability to take certain values.

(a) Explain the mathematical difference between continuous data and discrete data.

.....

..... **[2]**

(b) A person's exact foot length is measured in centimetres.
This measurement is then used to determine their shoe size, which can take values such as 7, 7.5, 8, or 8.5.

(i) State whether exact foot length is discrete or continuous data.

..... **[1]**

(ii) A student argues that shoe size must be continuous because it includes decimal values like 7.5.
Explain why the student is incorrect.

.....

.....
[2]

18 A factory packs identical books into boxes.
Each box contains exactly 24 books.

(a) State whether each of the following variables is discrete or continuous.

(i) The number of books in a box.

..... **[1]**

(ii) The mass of a book.

..... **[1]**

(b) The mass of one book is 850 g, correct to the nearest 10 g.
Find the upper bound for the total mass of the 24 books.
Give your answer in kilograms.

..... kg **[2]**

(c) An empty box has a mass of 1.2 kg, correct to 1 decimal place.
Calculate the lower bound for the total mass of a box containing 24 books.

..... kg **[3]**

1 **(a)** A and D **[B1]**

(b) B and C **[B1]**

[Back to Question] *[Total Marks: 2]*

2 **(a)** Continuous **[B1]**

(b) Discrete **[B1]**

(c) Continuous **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(iii) Continuous **[B1]**

(iv) Discrete **[B1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Discrete **[B1]**

(b) Continuous **[B1]**

(c) Continuous **[B1]**

(d) Continuous **[B1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** Discrete **[B1]**

(b) Continuous **[B1]**

(c) Continuous **[B1]**

(d) Discrete **[B1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(iii) Continuous **[B1]**

(iv) Discrete **[B1]**

(b) Width is measured (rather than counted) and can take any numerical value within a given interval **[B1]**

(c) A measurement of 6 metres would belong to both intervals (overlapping boundary) **[B1]**

$4 \leq w < 6$ (or $4 < w \leq 6$) **[B1]**

[Back to Question] *[Total Marks: 7]*

7 (a) Lower bound = 37.5 **[B1]**

Upper bound = 42.5 **[B1]**

(b) Smallest possible number = 1248 **[B1]**

Largest possible number = 1252 **[B1]**

(Note: Continuous boundaries are 1247.5 and 1252.5, but as a discrete integer variable, it must take the integer values strictly within this range.)

[Back to Question] *[Total Marks: 4]*

8 (a) Data Set A (time) is continuous, so it can take any decimal value and inequalities are needed to define exact boundary limits without gaps. **[B1]**

Data Set B (number of push-ups) is discrete, so it only takes integer values and a hyphen clearly defines the inclusive start and end integers. **[B1]**

(b) (i) 19.5 and 29.5 **[B1]**

(ii) 10 **[B1]**

(c) Age (usually given as an integer in years) OR Money/Currency (rounded to nearest cent/pence) OR Shoe size **[B1]**

[Back to Question] *[Total Marks: 5]*

9 (a) k is discrete **[B1]**

m is continuous **[B1]**

(b) For one rod, $1.95 \leq r < 2.05$, and for the total mass, $19.95 \leq m < 20.05$ **[B1]**

$1.95k < 20.05$ giving $k < \dfrac{401}{39} = 10\dfrac{11}{39}$ **[M1]**

$2.05k > 19.95$ giving $k > \dfrac{399}{41} = 9\dfrac{30}{41}$ **[M1]**

Since k is discrete, k must be an integer, so $k = 10$ **[A1]**

(c) (i) Mean of $d = \dfrac{0(24) + 1(15) + 2(7) + 3(3) + 4(1)}{50} = \dfrac{42}{50}$ **[M1]**

0.84 **[A1]**

d is discrete with exact individual values known, whereas L is continuous data grouped into intervals so individual rod lengths are unknown and mid-interval values must be used **[B1]**

(ii) $\dfrac{18 + 17}{50} \times 100 = 70\%$ **[B1]**

L is continuous data grouped in the interval $50.0 \leq L < 50.2$, so the exact distribution of the 17 lengths within that interval is unknown **[B1]**

[Back to Question] *[Total Marks: 11]*

10 **(a)** Continuous **[B1]**

(b) Height is continuous / can take decimal values. **[B1]**

A plant with a height between the intervals (e.g. 19.5 cm) would not fit into any class. **[B1]**

(c) Discrete **[B1]**

[Back to Question] *[Total Marks: 4]*

11 **(a)** Discrete data can only take specific or exact values (often counted) **[B1]**

Continuous data can take any value within a given range (often measured) **[B1]**

(b) **(i)** Continuous **[B1]**

(ii) Rounding limits the precision of the recording, but the actual true length could still be any value between the rounded boundaries **[B1]**

[Back to Question] *[Total Marks: 4]*

12 **(a)** Continuous **[B1]**

(b) Uses midpoints: 90, 110, 130, 150 **[M1]**

$\Sigma fx = 90 \times 12 + 110 \times 26 + 130 \times 15 + 150 \times 7 = 6940$ **[M1]**

$\frac{6940}{60} = 115.66... = 116$ or 115.7 **[A1]**

(c) The exact masses of the apples in each class are not known (or similar reference to data being grouped) **[B1]**

[Back to Question] *[Total Marks: 5]*

13 **(a)** Length **[B1]**

(b) **(i)** 14.45 **[B1]**

(ii) 14.55 **[B1]**

(c) Identifies that the discrete values include $5, 6, 7, 8, 9$ OR uses continuous boundaries $9.5 - 4.5$ **[M1]**

5 **[A1]**

[Back to Question] *[Total Marks: 5]*

14 **(a)** Continuous **[B1]**

(b) Identifies midpoints: $150, 250, 350, 450$ **[M1]**

15 **(a)** N is discrete (integer values only), so there are no values between 12 and 13 and inclusive inequalities ($10 \leq N \leq 12$, $13 \leq N \leq 15$) cover all counts without gaps **[B1]**

T is continuous, so strict upper inequalities ($48 \leq T < 50$, $50 \leq T < 52$) are needed to include all decimal values up to 50 without overlapping **[B1]**

(b) For N: $\sum fm = 6(11) + 14(14) + 12(17) + 8(20) = 626$ **[M1]**

Estimated mean of $N = \dfrac{626}{40} = 15.65$ **[A1]**

For T: $\sum fm = 5(49) + 15(51) + 13(53) + 7(55) = 2084$ **[M1]**

Estimated mean of $T = \dfrac{2084}{40} = 52.1$ seconds **[A1]**

(c) (i) The mean is a measure of average and does not have to be a possible value of a discrete variable **[B1]**

No, the student is incorrect; 15.65 is a valid mean even though N takes only integer values **[B1]**

(ii) Diagram for N: Bar chart (or vertical line graph) **[B1]**

Diagram for T: Histogram (or cumulative frequency curve / frequency polygon) **[B1]**

[Back to Question] *[Total Marks: 10]*

16 (a) (i) Discrete **[B1]**

(ii) Continuous **[B1]**

(b) The variable can still only take specific, fixed intervals/values (e.g., jumping by 0.5) **[B1]**

It cannot take any value within a range (e.g., a shoe size of 8.23 does not exist) **[B1]**

[Back to Question] *[Total Marks: 4]*

17 (a) Continuous data can take any value within a given range (measured). **[B1]**

Discrete data can only take specific, exact values (usually counted). **[B1]**

(b) (i) Continuous **[B1]**

(ii) Shoe size can only take specific, set values (there are gaps between sizes like 7 and 7.5). **[B1]**

It cannot take every possible value (e.g. you cannot have a shoe size of 7.23), so it must be discrete. **[B1]**

[Back to Question] *[Total Marks: 5]*

18 (a) (i) Discrete **[B1]**

(ii) Continuous **[B1]**

(b) Upper bound of mass of one book = 855 g **[M1]**

$24 \times 855 = 20520$ g $\implies$ 20.52 **[A1]**

(c) Lower bound of empty box = 1.15 kg **[B1]**

Lower bound of mass of one book = 845 g = 0.845 kg seen **[B1]**

Total lower bound $= 1.15 + (24 \times 0.845) = 21.43$ **[A1]**

[Back to Question] *[Total Marks: 7]*